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Technical Report

ROI Detection Module

v1.0

Comparative Analysis of YOLO Models for ROI Detection

Project Subject: Glaucoma detection using intelligent analysis of fundus images

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Contents

1 Introduction **2**

2 Dataset and Augmentation **2**

3 Evaluation Metrics **3**

4 Quantitative Results **3**

5 Training Curves Analysis **3**

5.1 YOLOv8 3
5.2 YOLOv9 4
5.3 YOLOv11 5
5.4 YOLOv12 6
5.5 YOLOv26 7

6 Comparative Analysis **8**

7 Justification for Selecting YOLOv9 **9**

8 Summary **9**

1 Introduction

The purpose of this report is to compare selected object detection models from the YOLO family in the task of detecting the region of interest, referred to as ROI, in fundus images used in the glaucoma detection project. In the proposed processing pipeline, the detection model acts as the first stage of image analysis. Its task is not to directly classify glaucoma, but to correctly locate the relevant image region that can later be passed to further analysis modules, such as a classification model or feature extraction stage.

The following models were compared:

- YOLOv8,
- YOLOv9,
- YOLOv11,
- YOLOv12,
- YOLOv26.

The models were evaluated using standard object detection metrics: mAP@50, precision, recall, and F1-score. Additionally, the training curves were analyzed, including training losses, validation losses, and detection quality metrics.

2 Dataset and Augmentation

The experiment was performed on a dataset that initially contained 1020 images. In order to increase the diversity of the training data and improve model generalization, data augmentation was applied. After augmentation, the training set contained 6100 images, while the test set contained 306 images.

The applied augmentation configuration included the following transformations:

- outputs per training example: 10,
- horizontal flip,
- rotation between $-15°$ and $+15°$,
- hue change between $-20°$ and $+20°$,
- saturation change between -25% and $+25\%$,
- brightness change between -24% and $+24\%$,
- blur up to 2.5 px,
- noise up to 1.4% of pixels.

The use of augmentation was important because fundus images may differ in terms of brightness, contrast, acquisition quality, noise level, and image orientation. By applying augmentation, the models should become less sensitive to small variations in the input images and should generalize better to unseen test samples.

3 Evaluation Metrics

The following metrics were used to evaluate the models:

- **Precision** – indicates what proportion of model detections were correct.
- **Recall** – indicates what proportion of actual objects were detected by the model.
- **F1-score** – the harmonic mean of precision and recall:

$$F1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall}$$

- **mAP@50** – mean average precision calculated at an IoU threshold of 0.50.

4 Quantitative Results

Table 1 presents the quantitative comparison of the evaluated YOLO models.

Table 1: Comparison of YOLO model performance

Model	**mAP@50**	**Precision**	**Recall**	**F1-score**
YOLOv8	97.0%	97.5%	96.5%	97.0%
YOLOv12	97.6%	97.5%	96.9%	97.2%
YOLOv11	**98.1%**	97.6%	97.1%	97.3%
YOLOv26	97.5%	**98.0%**	**98.0%**	**98.0%**
YOLOv9	**98.1%**	**98.0%**	97.0%	97.5%

Based on the numerical results, all models achieved high detection performance. The differences between the models are relatively small, but they are still relevant when selecting the final detector for the processing pipeline. The highest mAP@50 value was achieved by YOLOv9 and YOLOv11, both reaching 98.1%. The highest F1-score was obtained by YOLOv26. However, YOLOv9 provided a very strong compromise between mAP@50, precision, recall, and overall training stability.

5 Training Curves Analysis

5.1 YOLOv8

Figure 1 presents the training curves for YOLOv8.

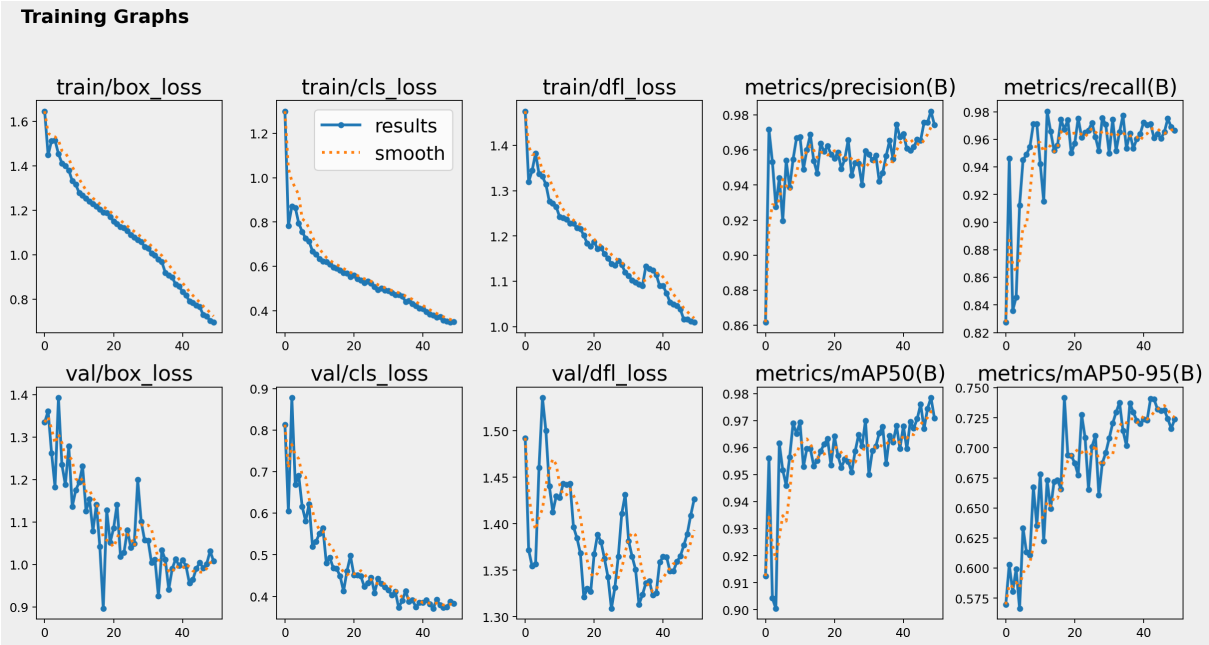

Figure 1: Training curves for YOLOv8

YOLOv8 shows a correct training process. The training losses, including *train/box_loss*, *train/cls_loss*, and *train/dfl_loss*, decrease systematically, which indicates that the model is effectively learning from the training data. Precision, recall, and mAP@50 reach high values after the first few epochs and then stabilize.

However, some fluctuations are visible in the validation metrics, especially for *val/dfl_loss* and mAP@50-95. This means that the model performs well, but it is not the most stable model among the compared variants.

5.2 YOLOv9

Figure 2 presents the training curves for YOLOv9.

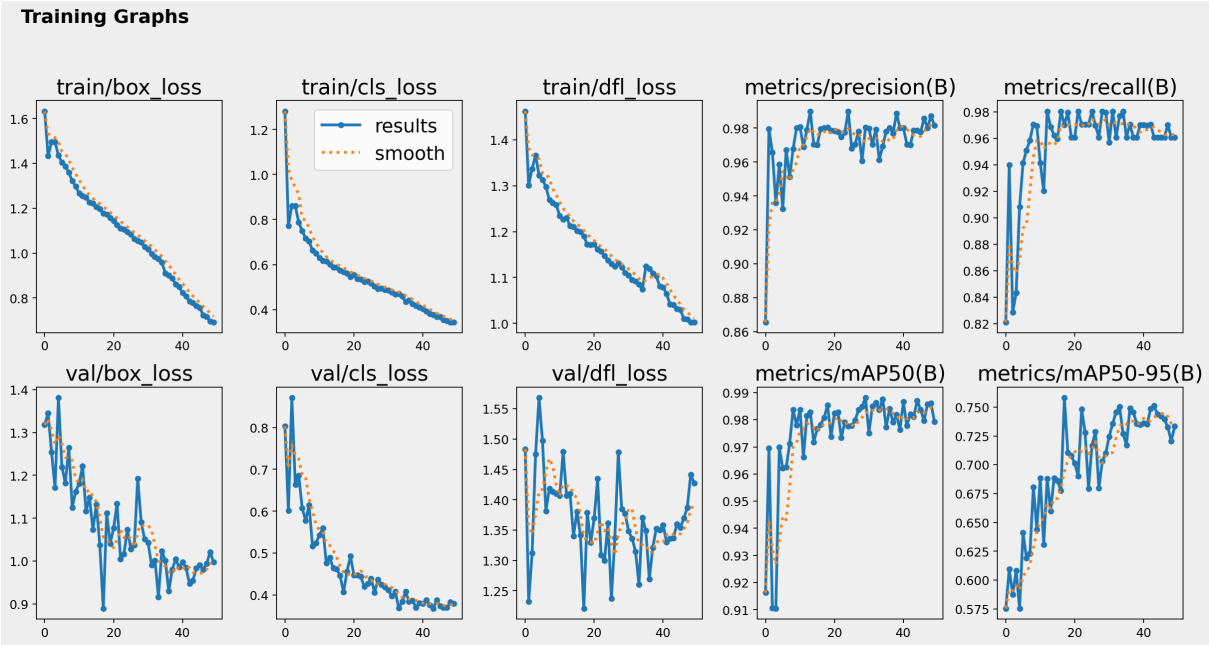

Figure 2: Training curves for YOLOv9

YOLOv9 achieved one of the best results in the entire experiment. The training losses decrease in a regular and consistent manner, while the detection metrics quickly reach high values. Precision remains close to 98%, recall is approximately 97%, and mAP@50 reaches 98.1%.

The curves also show that the model stabilizes after the initial training phase. An important observation is that YOLOv9 does not show a sudden degradation of validation metrics at the end of training, which suggests good generalization.

Compared with the other models, YOLOv9 provides a very favorable compromise between ROI localization accuracy, high precision, and stable learning behavior. This is especially important in a medical image analysis project, where an incorrectly selected ROI may affect the result of the subsequent glaucoma classification stage.

5.3 YOLOv11

Figure 3 presents the training curves for YOLOv11.

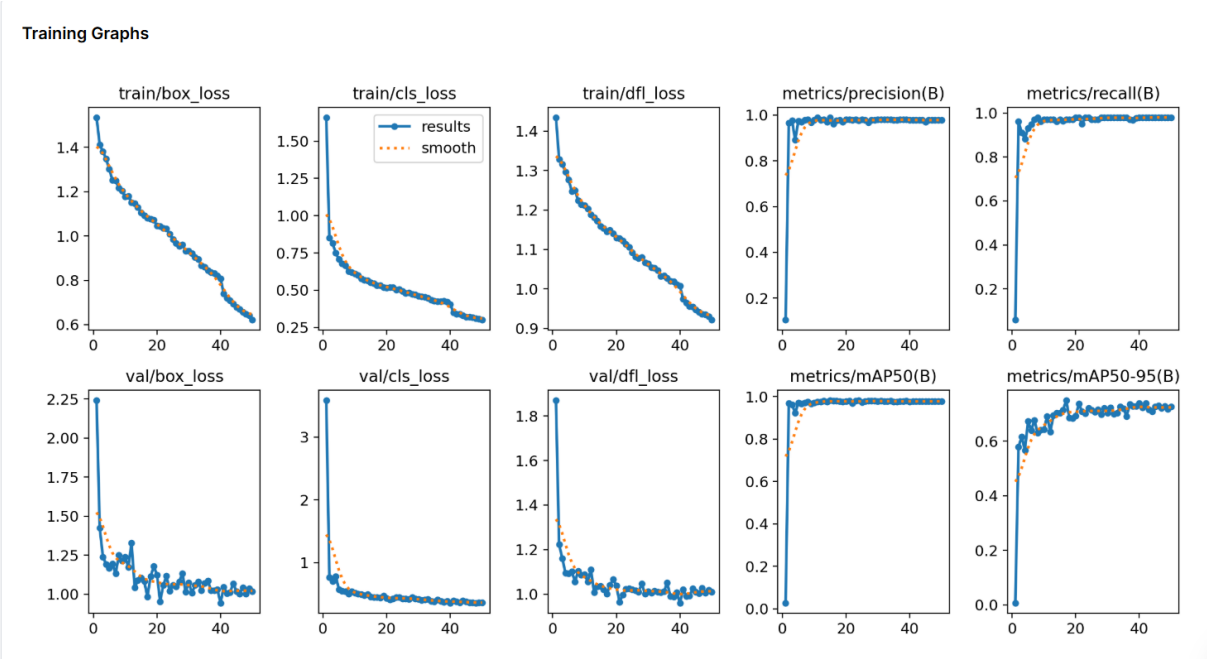

Figure 3: Training curves for YOLOv11

YOLOv11 also achieved a very high mAP@50 value of 98.1%. The model reached high values of precision and recall, equal to 97.6% and 97.1%, respectively. The curves show a decrease in both training and validation losses, although more visible fluctuations can be observed in the early validation phase.

Despite these fluctuations, the final performance metrics are very strong. YOLOv11 can therefore be considered a highly competitive model compared with YOLOv9. However, YOLOv9 achieved higher precision and a higher F1-score.

5.4 YOLOv12

Figure 4 presents the training curves for YOLOv12.

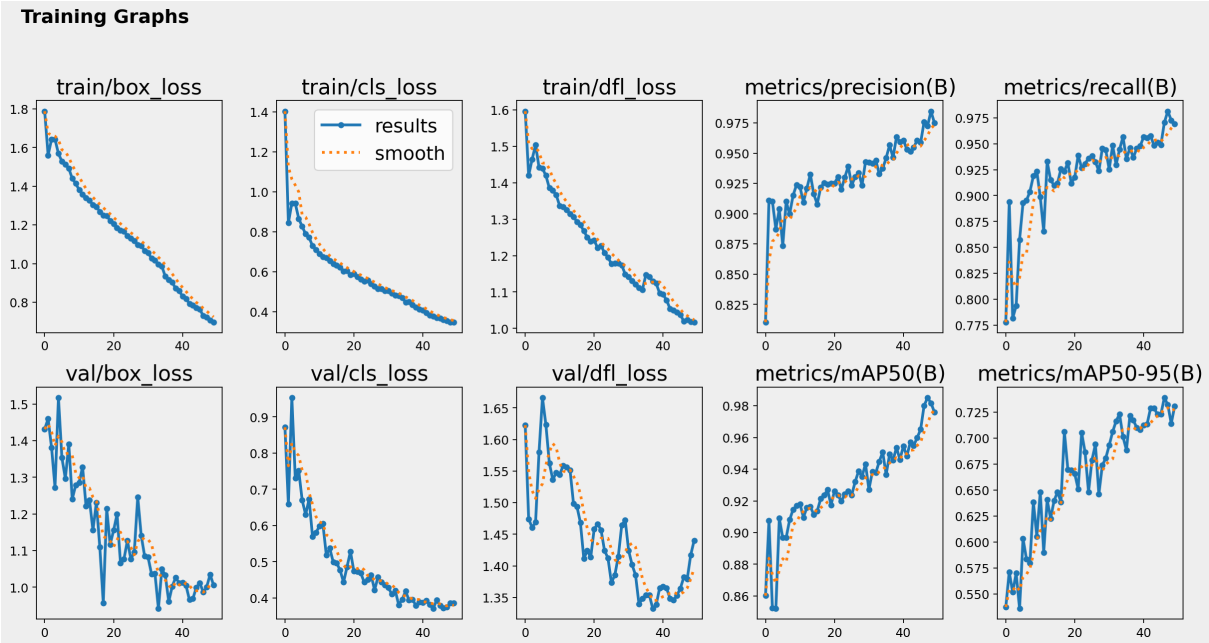

Figure 4: Training curves for YOLOv12

YOLOv12 achieved an mAP@50 value of 97.6%, precision of 97.5%, and recall of 96.9%. The model trained correctly, and the training losses decreased over subsequent epochs. Compared with YOLOv9 and YOLOv11, however, the final mAP@50 value was slightly lower.

The curves show a gradual increase in precision, recall, and mAP, but the model does not reach the same final effectiveness as the best-performing variants. YOLOv12 is a stable model, but in this experiment it was not the strongest candidate for the final pipeline.

5.5 YOLOv26

Figure 5 presents the training curves for YOLOv26.

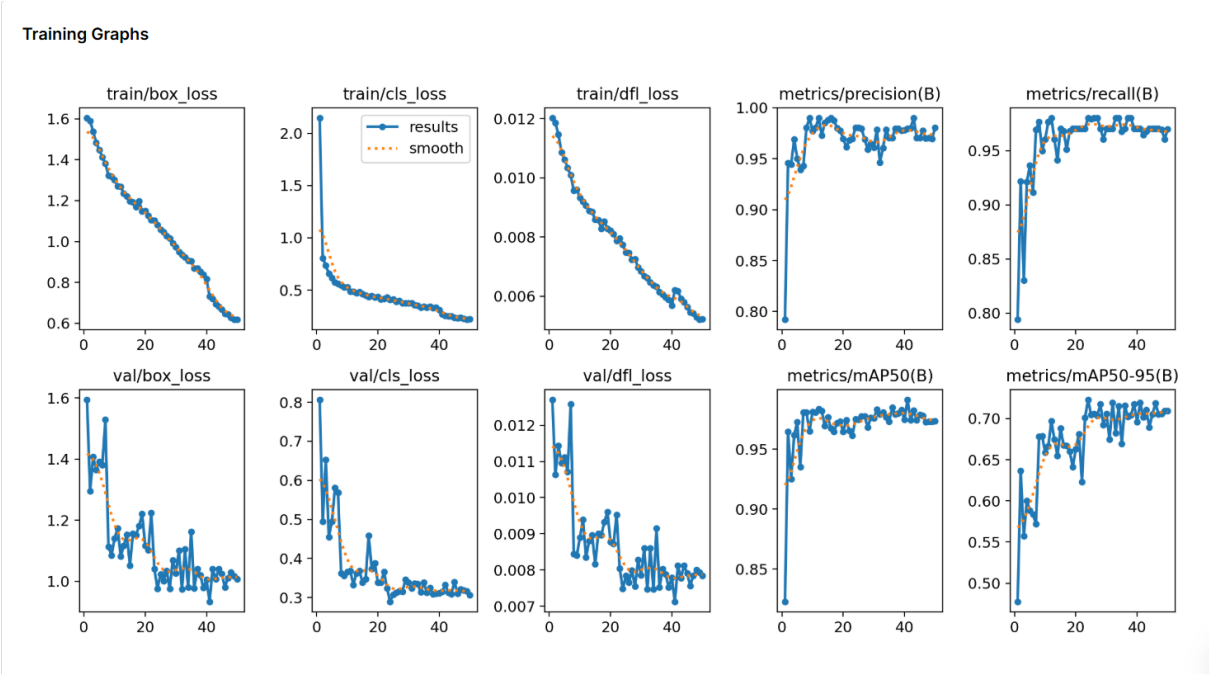

Figure 5: Training curves for YOLOv26

YOLOv26 achieved the highest F1-score, equal to 98%, as well as very high precision and recall values, both equal to 98%. This makes the model highly effective in terms of balancing correct detections and coverage of actual ROI regions.

The training curves show that YOLOv26 quickly reaches high values of precision, recall, and mAP@50. At the same time, its mAP@50 value was 97.5%, which is lower than the result achieved by YOLOv9 and YOLOv11. In the task of ROI detection, the F1-score is important, but mAP@50 better reflects the quality of bounding box localization at a defined IoU threshold. For this reason, despite its very high F1-score, YOLOv26 was not selected as the final model.

6 Comparative Analysis

All evaluated models achieved high performance, which indicates that the dataset and augmentation strategy allowed the models to effectively learn ROI localization. The training loss curves show a systematic decrease, which confirms that the models were learning meaningful representations of the target regions.

The most important observations from the training curves are as follows:

- all models show decreasing training losses, indicating a correct learning process;
- precision, recall, and mAP@50 reach high values, usually above 97%;
- YOLOv9 and YOLOv11 achieved the highest mAP@50, equal to 98.1%;
- YOLOv26 achieved the highest F1-score, but its mAP@50 was lower than that of YOLOv9;
- YOLOv8 and YOLOv12 achieved very good, but not the best, final results;

- some fluctuations are visible in the validation losses, which is expected with a relatively small test set and extensive data augmentation.

In the context of glaucoma detection, the ROI detector should be accurate, stable, and robust to image variability. An error at the ROI detection stage may cause the next module to receive an incomplete or incorrectly shifted image region, which may reduce the effectiveness of the entire system.

7 Justification for Selecting YOLOv9

YOLOv9 was selected as the final model for the ROI detection pipeline. This decision is based on several reasons.

First, YOLOv9 achieved the highest mAP@50 among the evaluated models, equal to 98.1%. YOLOv11 achieved the same mAP@50 value, but YOLOv9 obtained higher precision and a higher F1-score. This means that YOLOv9 provides very accurate ROI localization while also limiting the number of incorrect detections.

Second, YOLOv9 offers a strong balance between precision and recall. Precision at the level of 98% means that the vast majority of detections are correct. Recall at the level of 97% means that the model detects most of the actual ROI regions. In a medical image analysis project, this balance is important because both false detections and missed ROI regions may negatively influence later processing stages.

Third, the YOLOv9 training curves indicate a stable learning process. The training losses decrease consistently, while the validation metrics reach high values and stabilize in the final epochs. There is no clear sign of a strong degradation in validation quality at the end of training, which suggests that the model is not strongly overfitted.

Fourth, YOLOv9 achieves a better mAP@50 result than YOLOv26, even though YOLOv26 has a higher F1-score. In ROI localization, the quality of bounding box alignment with the actual target region is critical. Therefore, mAP@50 is a particularly important criterion. YOLOv9 combines very high mAP@50, high precision, and a strong F1-score, making it the most balanced choice for further use.

8 Summary

Based on the conducted comparison, all analyzed YOLO models can be considered suitable for ROI detection in fundus images used in the glaucoma detection project. However, YOLOv9 provided the best compromise between localization quality, detection precision, sensitivity, and training stability.

YOLOv9 will therefore be used as a component of the final processing pipeline. Its role will be to automatically determine the ROI, which will then be passed to the next stages of the system responsible for image analysis and glaucoma detection support.